

CODE OF PRACTICE

BATANGAS COFFEE FARMERS FEDERATION (BaCoFFed)

For

BATANGAS KAPENG BARAKO



ACKNOWLEDGEMENT

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Part I: Justification for the request for protection

Introduction

The Batangas Coffee Federation (BaCoffFed) commits to preserving and promoting the rich heritage of Batangas *kapeng barako*, a well-known variety of coffee cultivated in the province of Batangas for generations. The Technical Working Group (TWG) developed this Code of Practice to establish standards for the production, processing, and marketing of Batangas *kapeng barako* to ensure quality, authenticity, and sustainability. By setting clear guidelines, the TWG, which will become the Batangas Coffee Board (BCB), aims to protect the integrity of this unique coffee variety while supporting the local coffee-growing community, i.e., the Batangueño farmers.

This document outlines best practices for cultivation, harvesting, processing, and distribution, aligning with national and international quality standards. It also emphasizes environmental sustainability and fair-trade principles to ensure that the production of Batangas *kapeng barako* continues to benefit both farmers and consumers. Through this initiative, the BCB seeks to strengthen the reputation of Batangas as a premier coffee-producing region and safeguard the cultural and economic significance of *kapeng barako* for future generations.

What is the purpose of requesting protection? The request for protection of Batangas *kapeng barako* is essential to preserving the authenticity, heritage, and economic value of this coffee variant. *Batangas kapeng barako*, known for its bold flavor and rich aroma, is deeply embedded in the cultural identity of the Batangueños and the Filipinos. However, because of the increasing commercialization and the influx of foreign coffee varieties, genuine Batangas *kapeng barako* faces threats such as misrepresentation, adulteration, and declining cultivation.

By securing official protection through geographical indication (GI) status or other legal safeguards, the Batangas Coffee Board is aiming for the following:

1. **Certification of authenticity.** The Batangas Coffee Board wants to avoid the mislabeling of *kapeng barako* as similar to other coffee varieties. Batangueños should safeguard the unique identity and reputation of *kapeng barako*. Many coffee drinkers thought that the coffee they would purchase from any market in Batangas would be *kapeng barako* because of their lack of understanding and knowledge of the profile of this coffee variety. It is for this reason that they could no longer distinguish the authentic *kapeng barako* from those of a different variety. Since Batangas *kapeng barako* became famous due to its flavor profile, coffee producers from other provinces would falsely claim their coffee variety as *kapeng barako*, thereby damaging the flavorful character of the authentic Batangas *kapeng barako*. This Code of Practice safeguards the unique identity and reputation of the Batangas *kapeng barako*.
2. **Protection of Batangueño coffee farmers.** This request for protection aims to provide economic benefits to local growers by ensuring fair pricing, market access, and sustainable production practices. Many farmers in Batangas rely on *kapeng barako* as their primary source of income. However, they face challenges such as low market prices, competition from imported coffee, and fluctuating production yields. Supporting them through fair trade practices, direct purchasing, and

government-backed initiatives ensures earning a sustainable livelihood and improving their quality of life.

3. **Maintain quality standards.** The Batangas Coffee Board, through this Code of Practice, establishes and enforces strict production and processing guidelines to maintain the high-quality standards that Batangas *Kapeng barako* is known for. Because this coffee variety is known for its strong aroma and bold flavor, it is essential that this variety must be distinguished from other coffee variants. By supporting the local farmers, we help maintain the authenticity of this unique coffee and ensure that consumers enjoy a premium product that cannot be easily replicated elsewhere. Without proper support, local farmers are at risk of exploitation by large corporations or middlemen who buy their produce at unfair prices. Ensuring direct access to markets, providing technical assistance, and recognizing Batangas *kapeng barako* under legal protections like Geographical Indication status can help prevent commercial exploitation of *kapeng barako*.
4. **Promote sustainability.** It will also be the responsibility of the Batangas Coffee Board to encourage the farmers to practice environmentally responsible farming practices that will preserve soil health, biodiversity, and water resources. Supporting Batangas coffee farmers can promote sustainable agricultural practices. Many small-scale farmers use eco-friendly methods, including organic farming and shade-grown coffee cultivation, which help preserve biodiversity and maintain soil fertility. By investing in local coffee, consumers can encourage ethical and environmentally responsible farming.
5. **Enhance market competitiveness.** It is also essential that *kapeng barako* will become highly competitive in domestic and international coffee markets. This can be done by certifying the authenticity and the quality of the *kapeng barako*.

Protecting Batangas *kapeng barako* is vital for the sustainability of the local coffee industry and the preservation of a heritage product that holds deep historical and cultural significance. The Batangas Coffee Federation is dedicated to ensuring that this treasured coffee variety continues to thrive, benefiting farmers, consumers, and the broader coffee industry in local and international markets.

Reputation

This Code of Practice is designed to establish and uphold a reputation of **quality, authenticity, sustainability, and cultural integrity**. While it is gaining broader recognition, its emerging reputation is built on several key pillars:

1.Authenticity Shield. This Code of Practice serves as a standard yardstick for what qualifies as authentic Batangas *kapeng barako*. It protects the name and product from misrepresentation by setting strict standards on origin, variety, and production methods. This strengthens the coffee's identity and reputation both locally and internationally.

2.Quality Assurance Indicator. This Code places *kapeng barako* as a superior quality coffee by enforcing best practices from planting to post-harvest handling. This reputation

for high standards attracts conscious consumers and niche markets that value traceability and quality.

3. Model for sustainability and fair trade. Getting the assurance that the coffee production of *kapeng barako* follows an environmentally responsible farming practices and fair-trade principles in this Code of Practices enhances its reputation as an avant-garde ethical framework. It represents not just a product, but a sustainable livelihood for farmers and respect for the land.

4. Instrument for Geographical Indication readiness. The Code plays a foundational role in the application for Geographical Indication (GI) or other legal protection statuses. Its existence shows that the Batangas coffee industry is organized, committed to standards, and ready for national or international recognition, thereby boosting its credibility.

5. Means for cultural and economic advocacy. The Code is increasingly viewed as a way to revive and protect local agricultural heritage, especially as *kapeng barako* faces threats from market neglect and foreign varieties. It shows the coffee community's practical efforts to sustain its importance in modern markets.

The reputation of the Code of Practice for Kapeng Barako is a trusted, credible, and culturally rooted framework for protecting and promoting the most iconic coffee variety of the Philippines. It embodies a commitment to excellence, authenticity, and sustainable development, making it a critical tool in elevating the status of Batangas coffee both locally and globally.

History of Kapeng Barako

(An Excerpt from the book "Mapping Out the Batangueño Identity: The Batangas Kapeng Barako")

The evolution of coffee cultivation in the Philippines has undergone numerous shifts primarily due to economic demands, colonial influences, and changing agricultural conditions. The world recognized the Philippines as one of the leading producers of coffee globally during the early days. Today, there is a palpable desire to reclaim the former glory of Philippine coffee. Although the challenges are significant, Batangueños are working diligently to revive the *kapeng barako* industry because this legacy is worth preserving.

In the article written by Atienza (2024), she held that at the beginning of the 17th century, Spain was the most prosperous nation in Europe as it had a large colonial empire and holdings in Europe. Spain's wealth came from the gold and silver mines in its South American colonies. But in the mid-17th century, its power declined due to successive wars that weakened and drained Spain's resources. Spain tried to maintain its overseas empire until the early 19th century. To recapture lost glory and revive its prestige marked the spirited attempt to stimulate economic development in the Philippines, thus, triggering economic recovery. This led to the establishment of the coffee industry.

Analyzing the historical narrative regarding coffee reveals conflicting accounts about its introduction to the Philippines. Kape de Filipina (2023) might have referenced Rafael Bartolome (1962) when it claims that in 1740, a Spanish friar named Claudio de la Concepcion brought three *gantas* of coffee, allegedly instructing a servant to plant these seeds in a garden located in Laguna or Lipa; later, they planted them on his father's land in Pinagtungulan, Lipa (Atienza).

However, Café de Lipa argues that two Franciscan friars cultivated coffee, and they taught two Filipino natives the technique of propagating coffee plants. Although these individuals were identified as members of the Macasaet family, who significantly contributed to the growth and care of the Liberica seedlings, this divergence in historical narratives highlights the complexities surrounding the origins of coffee in the Philippines.

According to historical records, the introduction of the coffee variant known as *Coffea liberica* to the Philippines occurred later. The variety that was brought to the Philippines by the Franciscans was the *Coffea arabica*. The Augustinians, who took care of the spiritual life of the Batangueños, were responsible in cultivating coffee in Lipa.

The Bureau of Plant Industry indicates it was in Barrio Pinagtungulan, Lipa, where they cultivated many plant seedlings. The Macasaet clan asserts that their forefathers were responsible for sowing the first coffee seeds. However, a historian from Lipa contends that the town's Captain, Francisco Mantuano, introduced, in 1784, the inaugural coffee seeds to Lipa, which he subsequently planted in his backyard; this claim complicates the narrative. Although the Macasaet clan's assertion holds cultural significance, it is essential to consider multiple perspectives in historical discourse (Villa, 2002).

Consequently, an apparent lack of documentation arises regarding the plantation and cultivation of coffee trees. Historians suggest that this may be due to the demise of these trees, which is most likely connected to the volcanic eruptions. Indeed, the historical records indicate that the Taal volcano experienced eruptions in 1749, 1754, 1781, 1790, and 1808; however, this absence of records raises questions about the impact of such events on local agriculture. Although the timeline is clear, the reasons for the documentation gap remain elusive because of the natural disasters that struck the region.

In 1778, Governor General Jose de Basco y Vargas (1733-1805) arrived in the Philippines to fulfil his role as the governor of the archipelago under the auspices of the Spanish Empire. In his term as governor-general, people recognized him as one of the most economically astute governors-general to have governed the Spanish-ruled Philippines. On March 10, 1785, the government instituted the *Real Compañía de Filipinas* (The Royal Company of the Philippines) through a royal decree (Alchetron, 2024), and this Royal Company was chartered entity designed to establish a trade monopoly throughout the Spanish territories in the Philippines.

The Enterprise aimed to forge commercial ties among the various colonies and between these territories and Spain. It aimed to supply Manila with European products while simultaneously facilitating the export of not only Philippine goods but also merchandise sourced from the Orient. Furthermore, an equally significant aim was to promote agricultural development within the Philippines. As the coffee export market flourished, numerous farmers, particularly in Lipa, opted to cultivate and propagate coffee (The Kahimyang Project, 2024).

To enhance the visibility of the coffee industry, the Gobernadorcillo, Don Gallo de los Reyes, initiated a campaign that underscored the significance of coffee cultivation. He mandated that every household must plant a coffee tree in their backyard. This order remained in effect until 1832, when his son, Santiago de los Reyes, was elected to the same post. Santiago asserted that those who failed to comply with the order would receive beatings and had to walk around the town with an offending sign on their back (Atienza).

In 1814, due to a vigorous influx of agricultural commodities into the global market, and perhaps also because the Augustinian priests, particularly Fathers Elias Nebreda and Benito Varas, were acutely aware of coffee's burgeoning reputation in international trade, they actively encourage the local populace to engage in coffee cultivation. This initiative subsequently facilitated the dissemination of coffee cultivation practices to other places within Batangas, including Ibaan, Lemery, San Jose, Taal, and Tanauan. The plant, botanically classified as *Coffea arabica*, flourished in Batangas' tropical climate and volcanic soil.

Coffea arabica spread rapidly and, similar to developments in various other nations, it became a highly valued agricultural commodity among Filipino cultivators. The soil and climatic conditions of Batangas, which were exceptionally favorable, established it as a crucial center for coffee cultivation within the archipelago. The humble seedling signaled the beginning of a coffee revolution in the Philippines. The climate and topography of Lipa proved to be particularly advantageous for coffee production; this led to its swift transformation into a thriving industry. By that point, coffee became remarkably prevalent among Batangueños. As a result, *Coffea arabica* earned the title of "*Kapeng Tagalog*," indicating its indigenous roots. The *cabeza de barangay* of Lipa, known as Francisco Montuano, amassed significant wealth because of the extensive cultivation of coffee.

Although the eruptions of Taal volcano significantly impacted the coffee industry, coffee farmers continued with great determination in their cultivation efforts. This determination of the coffee farmers led to the growth of the coffee industry, especially when the Philippines adopted Trade Liberalization, a direct result of the cessation of the galleon trade between Mexico, Canton, and Manila. As a result, this unassuming seedling marked the beginning of a coffee revolution in the country since the region has a climate and terrain suited for coffee cultivation. Eventually, the coffee business developed into a thriving industry. In the 1800s, the Philippines has emerged as the fourth leading coffee-producing and exporting nation, celebrated for its rich and aromatic beans. Nevertheless, it was the resilience of these farmers that characterized this period.

Following its promising debut in Batangas, the cultivation of coffee spread to various regions across the Philippines notably, including Cavite, Laguna, and Mindanao, and the elevated terrains of Benguet in Luzon. By the advent of the 19th century, coffee (notably *Coffea arabica*) had established itself as a popular crop among Filipino agrarians. However, by the midpoint of the 19th century, the increasing demand for coffee in the international market was becoming evident, and the Philippines emerged as a crucial producer of this commodity. This nation's favorable positioning, integral to Spanish-controlled trade routes, facilitated its integration into the global coffee market. Thus, coffee became a significant export product directed toward the United States and Europe because of its increasing popularity and demand. Although challenges existed, the industry flourished.

The illustrious coffee era in Lipa spanned from 1840 to 1890, leading to the remarkable growth and expansion of Batangas coffee within the global market despite the fluctuating supply of this esteemed beverage (Atienza). Beginning in 1860, the coffee trees that endured the volcanic eruptions continued to thrive and produced an abundance of cherries. The volcanic ash appeared to serve as a fertilizer, which enabled the coffee trees to flourish even more, ultimately elevating Batangas as the fourth-largest coffee supplier worldwide. In 1869, Batangas emerged as the epicenter of coffee in the Philippines. It also became the primary supplier of coffee to the United States under the brand name “Batango” coffee, which commanded the highest prices among various coffees sold in neighboring countries. Furthermore, the opening of the Suez Canal in that same year, coupled with the American West’s expansion after the Civil war concluded in 1865, significantly opened up vast European and American markets for Philippine agricultural products (Atienza).

The growing demand for coffee prompted Captain Crispin Farol of Batangas to initiate a program fostering economic growth. He commenced his initiative with local farmers. He mandated that those who failed to comply with the directive to plant fruit-bearing crops, particularly coffee, which showed significant promise for economic growth, would face severe penalties. Between 1869 and 1871, the coffee sector flourished in Batangas, and Don Celestino Solis gained recognition as the largest coffee producer in the region (Encarnacion, 2002).

By the 1880s, the Philippine archipelago was experiencing a notable coffee boom. The town of Lipa emerged as the nucleus of the Philippine coffee industry. Thus, it earned the title “Coffee Capital of the Philippines.” This label was due to its fast coffee plantations and substantial contributions to the national economy. During this time, the Philippines rose to become the fourth-largest coffee producer in the world and at one point, it claimed the title of the exclusive coffee supplier to the United States.

The coffee industry not only flourished but also helped the accumulation of wealth for numerous Batangueño farmers. This developing sector catalyzed significant economic growth. Because of this, the political leaders forced the people to cultivate coffee and other agricultural products. During this period, Kapitan Julian Rosales distributed seedlings of coffee. Don Vicente Agregado, who represented the second district of Batangas at the time, also played a crucial role in revitalizing the development of agricultural products. During this time, Lipa earned the title “Coffee Capital of the Philippines.” Although other regions of the world—particularly South American countries—suffered from coffee rust, the coffee industry in Batangas thrived, ultimately accounting for 7% of the total Philippine exports.

Due to Lipa’s economic growth, which reached approximately 4 million pesos, Queen Regent Maria Cristina of Spain issued a decree granting Lipa the title Villa de Lipa. This title allowed the city to use the Coat of Arms facilitated through Don Victor Balaguer (Kasaysayan ng Lipa, 2022).

The Decline of the Coffee Industry in the Late 19th Century

The Philippine coffee industry underwent a fleeting phase of prosperity; however, the coffee industry declined because, during the latter years of the 1880s, the country suffered a disastrous epidemic of *Hemileia vastatrix* (commonly known as coffee leaf rust) along with an invasion of coffee borers. These insects cause substantial harm to coffee

plants. This specific coffee affliction ravaged a significant portion of the *arabica* crop, recognized as *kapeng Tagalog*, particularly in the province of Batangas. Furthermore, the emergence of fungi and the genus *Aphis* severely weakened the entire coffee sector. This calamity transpired between 1890 and 1894, just before the Philippine-American Revolution.

Brazil emerged as a formidable force in the coffee sector, swiftly ascending to become the world's largest coffee producer. Many Batangueño farmers abandoned coffee cultivation in favor of alternative crops, such as sugarcane, citrus, and rice. Although this transition was due to economic reasons, it ultimately diminished the Philippines' prominence in the coffee trade.

Despite the significant downturn experienced by the Philippine coffee industry in the latter part of the 19th century, there was a subsequent recovery in the early 20th century. During this period of transformation, the coffee aficionados introduced the *Coffea canephora* (robusta) variety. These cultivars exhibited higher resistance to coffee rust and could thrive at lower elevations. This adaptability rendered them particularly well-suited to the Philippine climate.

The Discovery of the Coffea liberica

The coffee industry thrives globally, illustrated by the *Coffea arabica*, often regarded as the first variety cultivated in Lipa, and *Coffea canephora* (robusta). However, another coffee variety debuted in 1843 within the tropical forests of West and Central Africa, specifically Liberia, designated as *Coffea liberica*. This particular variety flourished naturally in Sierra Leone, Ivory Coast, and Ghana (Spring, 2024).

Although *arabica* and *robusta* had gained widespread acclaim worldwide, *liberica* was not readily accepted by coffee enthusiasts, except those seeking unique and diverse flavor profiles. This scenario unfolded against an increasing global demand for coffee, which the current supply could not sufficiently satisfy. The dependence on *robusta* alone proved inadequate due to the declining *arabica* stocks further strained by coffee rust. At this time, *Coffea liberica* was introduced into the international coffee market, acting as a solution to the challenges faced by other coffee varieties.

European colonial powers (including the Dutch and the British) began cultivating *liberica* coffee in their colonies, particularly in Southeast Asia, like the Philippines, Indonesia, and Malaysia, and also in the Caribbean. However, because *liberica* shows greater resilience to diseases and pests, it became a viable crop in areas where *arabica* faced challenges. Because it thrives in lowland tropical climates marked by high humidity, this distinct adaptability makes it a valuable alternative to other coffee species.

Coffea liberica was introduced to the Philippines by American colonizers between 1890 and 1894, just before the onset of the Filipino-American Revolution. This specific variety of coffee found an optimal environment in Batangas; however, Manuel Genato (Agoncillo) was the first to bring the *Coffea liberica* variety to Batangas from Manila in 1891. He cultivated three hectares of land in sitio Abra in Banay-Banay, San Jose, and Lipa. Furthermore, he distributed coffee seeds to the towns of Rosario in Batangas, San Pablo in Laguna, and Tiaong and Sariaya in Quezon (Morada, 1925).

As a result, the Batangueño coffee farmers often referred to it as a *kapeng Amerikano* (likely because it was the Americans who introduced this variety to the region);

this later transformed into the name *kapeng barako* (homophonic), derived from the *Coffea liberica* (kape [li]berica) variety. *Kape liberica* traditionally adopted the term *kapeng barako*. However, although this particular variety flourished and became a cultural emblem within Batangas—particularly in Lipa and nearby municipalities—it gained notoriety mainly due to its bold and robust flavor, which distinctly distinguished it from other types of coffee.

However, despite its manifold advantages, *Coffea liberica* encountered a series of challenges, including its relatively low yields. This occurs because *liberica* plants yield a fewer number of beans per hectare in comparison to both *arabica* and *robusta* cultivars, thereby rendering them less economically viable for large-scale production. Although the *liberica* coffee is characterized by a robust, smoky, and somewhat bitter flavor profile, which might not resonate with most consumers, this limitation restricts its demand within global coffee markets, predominantly influenced by milder *arabica* blends. Additionally, coffee farmers observed that due to the larger size of *liberica* beans compared to other coffee varieties, distinct processing techniques are needed, thereby introducing additional complexities for producers.

Notwithstanding, a segment of coffee aficionados recognized the merits of *liberica* due to its potent aroma and flavor; thus, it emerged as the preeminent choice among the Batangueños. In 1905, this audacious and robust coffee, characterized by its earthy and woody undertones, became popularly referred to as Batango coffee or Batangas coffee. It flourished particularly within the coffee's melting pot in America, especially in the bustling coffee market at Pike Place, Seattle, Washington, where the renowned coffee establishment Starbucks first took root.

Coffee During the American Colonial Period and World War II

When the *liberica* coffee first penetrated the United States market, it subsequently emerged as a favored variety within the Philippines, ultimately evolving into a cultural emblem in Batangas and later, in Cavite. *Liberica* continues to possess a distinctive place in Filipino coffee culture to this day.

In 1922, the government authorities distributed more than 60,000 seedlings (of both *liberica* and *excelsa* varieties) among the coffee cultivators of Batangas. The Provincial Inspector Ambrocio Lontok and Municipal Inspector Crispulo Bagui spearheaded this initiative; however, the introduction of these extra trees contributed to a greater yield of coffee in the region.

The Japanese occupation throughout World War II engendered significant disruptions in coffee production and trade. Numerous plantations fell into neglect and resources were redirected to fulfill various wartime exigences. This diversion inevitably culminated in a marked decline in coffee production during this tumultuous period. Simultaneously, when the Japanese commandeered international trade routes, coffee exports from the Philippines were effectively halted. Instead of prioritizing coffee cultivation, agricultural efforts were redirected towards local consumption or appropriated by Japanese authorities to bolster their military operations.

During the war, coffee ascended to the status of a luxury item for a considerable number of Filipinos, primarily due to its scarcity. Alternative beverages and substitutes emerged, including roasted rice, corn, or sweet potatoes, which were ingeniously brewed

to simulate coffee. Those fortunate enough to possess access to *kapeng barako* often resorted to stretching their supply by reusing grounds or diluting the brew.

However, in the light of the prevailing hardships that were quite significant, coffee maintained its cultural importance throughout the war. Sharing a cup of coffee—regardless of whether it was a substitute version—became an essential means for individuals to uphold a semblance of normalcy and community amidst trying times. Coffee, therefore, continued to be a powerful symbol of hospitality, but scarcity often necessitated the serving of reduced portions or alternative iterations to guests. This dynamic coffee appreciation illustrates the complex relationship between tradition and adversity, because, despite the challenges faced, the act of sharing coffee persisted.

Post-War Recovery and Coffee Revitalization

In the aftermath of the war, the Philippines encountered considerable obstacles in reconstructing the agricultural sector, particularly with regard to the coffee industry. Numerous plantations, which had previously thrived, were obliterated and the nation grappled with the arduous task of reinstating pre-war production levels. Efforts aimed at rehabilitating coffee cultivation encompassed the promotion of varieties such as *kapeng barako* and *robusta*. These varieties had demonstrated superior resilience under the exigencies of wartime conditions. However, to avert future crises, it became evident to the populace that a transition towards more sustainable agricultural practices and diversification was imperative.

The challenges encountered by coffee farmers and their communities during the war years not only underscored their resourcefulness, but also contributed significantly to the resilience and adaptability inherent in Batangueño coffee culture. Indeed, the scarcity of coffee experienced during World War II illuminated the tenacity and perseverance of these farmers when confronted with adversity.

In the decades spanning the 1950s and 1960s, significant endeavors were undertaken to reinvigorate the Philippine coffee sector, particularly in light of the growing global demand for coffee. International interest in *robusta*, especially for its application in instant coffee production, facilitated the Philippines' discovery of a niche market for its coffee offerings. During this period, the archipelago was already home to four predominant varieties of coffee: *arabica*, *robusta*, *excelsa* and *kapeng barako (liberica)*. Each of these varieties is cultivated in distinct regions across the nation, with specific locales specializing in particular cultivars. The principal coffee-producing regions in the Philippines at present encompass Batangas (*kapeng barako* or *liberica*), Cavite (*robusta*) and the Cordilleras, which includes Benguet and Sagada (*arabica*), as well as Bukidnon and Davao in Mindanao (*arabica* and *robusta*). However, the complex interplay of local traditions and global market trends has shaped the industry's evolution, making it a fascinating study of agricultural adaptation and economic resilience.

In the 1970s, Nestlé, a multinational corporation, engaged in a partnership with local farmers to facilitate training initiatives and promote large-scale *robusta* coffee production, intended for incorporation into instant coffee blends. The Philippine Coffee Board was subsequently established to bolster the industry and advocate for coffee as a viable agricultural crop for local farmers. During this period, specifically in 1970, numerous Lipeños endeavored to sustain the propagation of *kapeng barako*. Dr. Amado Silva, for

instance, cultivated coffee trees in the barrio known as Bayorbor, located in San Sebastian, Mataasnakahoy. Mr. Jose Mercado, a pivotal figure from the Macasaet clan, who played a significant role in the establishment of *Café de Lipa*, also contributed substantially to the propagation of *liberica*. Retired Col. Necitas “Katy” Katigbak dedicated portions of his land to the *liberica* coffee plantation, which, as of this date, maintains a considerable plantation in Brgy. Tipakan, Lipa City. He mandated that each household plant five trees in their backyards. Furthermore, he organized training activities for students in Grades 11 and 12, focusing on coffee farming at the Lipa Coffee Academy.

In the year 1977, the inaugural Batangas coffee brewery was founded (it bore the name of Siete Baracos Coffee). This establishment emerged as a pivotal entity, responsible for distributing packs of commercial coffee to the prominent markets of Manila and Cebu. However, this achievement was not merely a stroke of luck; it resulted from the unwavering patience and diligent efforts of Dr. Amado Silva (Cantos, 2013). Although the brewery faced numerous challenges, its impact on the coffee industry was undeniable because of its dedication to quality and service. This, in turn, helped to shape the coffee culture in the region.

Challenges in the 1980s and Decline in Production

During the decade of the 1980s, a significant decline in coffee production was observed. The global surplus of coffee precipitated a dramatic decrease in prices; however, this downturn rendered coffee farming increasingly unprofitable. As a consequence, numerous Filipino farmers abandoned coffee cultivation, culminating in a marked reduction in production levels. Although many farmers turned to alternative agricultural products because of the volatile pricing of coffee, the Philippine coffee industry encountered substantial setbacks. Because there were about 250,000 farmers who were depending on the coffee industry, the Republic Act 2712 was promulgated to prevent coffee imports so as to protect local producers (Agribusiness Weekly).

The nation's coffee production plummeted to unprecedented lows. To combat this decline, there has been a concerted effort to revitalize the coffee sector through various initiatives, including government programs, farmer cooperatives and collaborations with international organizations. Given the persistent high demand for coffee, the government exerted considerable effort to rejuvenate the industry. For this reason, the Philippine Coffee Roadmap (2017-2022) was implemented with the intent to promote sustainable coffee production, augment yields, and enhance the quality of Filipino coffee; thereby enabling it to compete effectively in the global market.

The coffee sector has, without a doubt, undergone a significant renaissance, primarily as a consequence of the burgeoning interest in specialty coffee and the populace's increasing affinity for local varieties. Consequently, both local and international demand for unique types, such as *kapeng barako*, *excelsa* and the *arabica* cultivated in elevated regions like Benguet and Sagada, has markedly escalated. Organizations including the Philippine Coffee Board—alongside various local NGOs—have collaborated with governmental entities to advance sustainable coffee farming practices, provide essential training and rejuvenate coffee production.

In the Province of Batangas, the Batangas Provincial Agriculturist Office, operating under the auspices of Governor Hermilando I. Mandanas, has instituted the Provincial Technical Working Group for Coffee. This initiative aims to propagate and promote the

coffee culture specific to Batangas; however, it also seeks to re-establish the cultural nexus between the Batangueño people and their cherished *kapeng barako*. Although the challenges inherent in this endeavor are significant, the collective effort demonstrates a commitment to elevating the region's coffee heritage.

Physical link to the territory

A. Topography and Slope

Batangas Province’s terrain is mostly composed of elevated lands, small low flat lands, rolling hills and numerous mountains. Elevated lands can be found in different municipalities of Malvar, Mataasnakahoy, San Jose and the cities of Lipa, Tanauan and Sto. Tomas. Lowland rice culture relatively can adapt in small low flat lands in the municipalities of Padre Garcia and some portions of Rosario, Taysan, Lemery, Calaca, Lian, San Juan and Nasugbu. Lands that are good for production forest and considered as rolling hills are found in Rosario, Lemery, Calaca, Lian, San Juan, San Luis, Tuy and Agoncillo. The land topography of Cuenca, Calatagan, Balayan, Ibaan, Bauan, San Pascual, Taal, San Nicolas, Sta. Teresita, Alitagtag and Batangas City are gently rolling with some mountains. They are often utilized for coconut plantations and agro-forestry development and the like.

Some parts of mountainous municipalities of Lobo, Mabini, Tingloy, Laurel and Talisay are considered protected areas. Furthermore, prominent mountains in the province are Lobo Mountains in Lobo, Batulao in Nasugbu, Panay in Mabini, Makulot in Cuenca, Banoy and Pinamucan in Batangas City, Mt. Malarayat Forest Reserve and Susong Dalaga in Lipa City, San Pedrino in Southwestern Balayan, Los Picos and Pico de Loro in Nasugbu and Compradia located between San Juan and Rosario. Batangas shares with Laguna and Quezon, with Mt. Makiling, and Mt. Malipunyo, and Tagaytay Ridge with Tagaytay City. Generally, Batangas has a rolling terrain with almost fifty percent (50%) of its land area having a grade of less than 15°. Slopes can be found along the shores of Verde Island Passage and the shores of Taal Lake. The central part of the province is Mt. Makulot with a summit of 957 meters above sea level.

Shown below is a detailed table of the distribution of Provincial Land Resources on Crop Suitability based on slope categories.

Table __. Slope Categories, Area, Description and Crop Suitability

Range (%)	Area (ha.)	Description	Crop Suitability
0 – 3	25,253	Level to nearly level	Irrigated/Paddy Fishpond
3 – 8	54,624	Gently sloping to undulating	Annual cultivated crops
8 – 18	8,490	Moderately sloppy to rolling	Perennial tree & vine crops
18 – 30	12,810	Strongly sloping to moderately steep	Plantation forest
30 – 50	17,750	Steep hills and mountains	Production forest

More than 50	87,654	Very steep hills	Forest reserve mountains
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Socio Economic and Physical Profile (SEPP), 2020

Batangas Province’s Slope Map designates what municipalities are level to gently sloping, gently sloping to undulating, undulating to rolling, rolling to hilly, hilly to steepy hilly and steepy hilly to mountainous.

Irrigated Land - Land serviced by natural irrigation or man - made irrigation facilities. These include lands where water is not readily available as existing irrigation facilities need rehabilitation or upgrading or where irrigation water is not available year-round.

Fishpond - A land-based facility enclosed with earthen or stone material to impound water for growing fish.

Annual Cultivated Crops - These are production areas devoted to sugarcane, corn, peanuts, and vegetables like eggplants, squash, mustard and root crops such as cassava, gabi, ube, and sweet potato.

Perennial Tree & Vine Crops - Plant or trees that last for more than two growing seasons, either dying back after each season or growing continuously.

Plantation forest - Generally intended for the production of timber and pulpwood increase the total area of forest worldwide.

Production forest - Production forests are forestlands utilized for timber production and non-timber uses. This can either be grassland, secondary forest or logged over areas which includes tenured forest land wherein utilization of forest resources in the area are regulated and well managed.

Forest Reserve Mountains - Forest lands that have been reserved by the President of the Philippines for any specific purpose or purposes.

As shown in the Table, major areas that are found to be Suitable to Irrigated Rice (SIR) land use are located in the municipalities of Balayan, Lemery, Lian, Nasugbu, Rosario, San Juan and Taysan, while minor areas are found in Lobo, Padre Garcia, Calatagan, Laurel and Calaca. These are low lying areas where irrigation water is abundant and can sustain rice crop production, if properly developed and maintained.

Municipalities that are usually found near the seashore or adjacent to river areas are considered to be suitable to Fishpond Production. These are the areas from Balayan, Calatagan, Lian, Lobo, Nasugbu, San Juan and Batangas City.

Moreover, the largest land area found Suitable to annual Cultivated Crops (SCC) which are generally devoted and planted to sugarcane crops can be found in western part of Batangas and a few number of hectares are situated in the eastern part of the Province.

Perennial Trees and Vines (PTV) production is identified and found to be suitable in Balete, Bauan, Cuenca, Malvar, Rosario, San Jose, San Pascual, and Cities of Tanauan, Batangas and Lipa.

There are minimal areas that are found to be suitable for pasture land purposes because of less availability of grass during the dry season. These are located at Calatagan and Tuy which are intermittently distributed by patches somewhere in Lipa City.

Furthermore, areas found to be suitable for forest plantation in the province are usually found and located at Laurel, Balayan, Lemery, Lian, Nasugbu, Lobo, Talisay, Agoncillo, Bauan, Mabini, Malvar, Rosario, San Juan, San Luis, Tingloy and Cities of Tanauan, Batangas and Lipa. Aside from plantation forest, areas suitable to production forest purposely for the production of timber are frequently found in Agoncillo, Alitagtag, Bauan, Lemery, Lobo, Mabini, Mataasnakahoy, San Jose, San Juan, San Luis, Taysan, Tingloy, Tuy Tingloy and Cities of Tanauan, Batangas and Lipa.

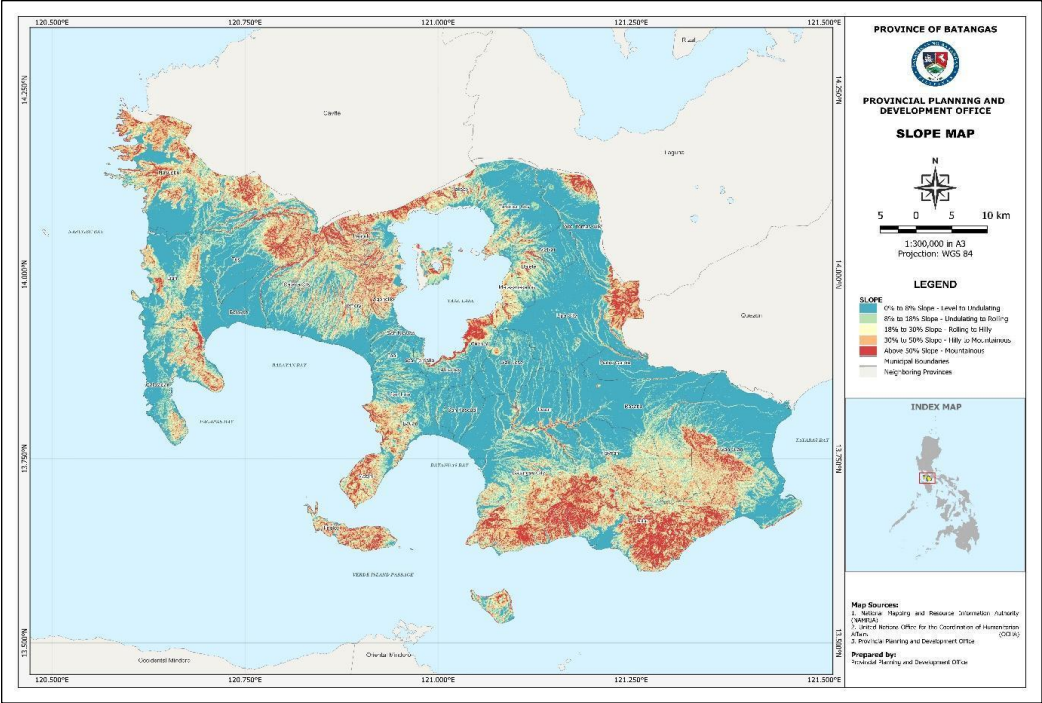


Figure __. Slope Map

B. Soil Physiography

Batangas Province is endowed with rich, volcanic soil above par in quality and easy to cultivate. The geographical formation of soil consists mostly of water-laid volcanic tuff. The tuffaceous formation covers the agricultural regions.

The characterization and classification of the different types of soils are according to the landforms and landscapes and delineated in the maps by Bureau of Soils and Water Management (BSWM) under the land resources evaluation project.

Table __. Batangas Major Land Form and Uses

Land Forms	Municipalities / Cities Commonly Covered	Present Use	Crop / Land Use Suitability
Lowland-Wetland	Nasugbu, Lian, Calatagan, Lobo & San Juan	Fishpond, Mangrove, Nipa	Fishpond
			Mangrove

			Nipa
Lowland Alluvial	Balayan, Lemery, Laurel, Balete, San Juan, Talisay	Rice, Sugar diversified	Rice and Annual Crops
	Santo Tomas, Tanauan City, San Juan, Lipa City, Balayan & Nasugbu	Rice, Sugar, Coco tree	Annual Crops
Lowland-Residual		Diversified Crops	
	Calaca, Sto. Tomas, Tuy & Balayan		
Upland			Perennial trees & Vine crops

Source: Batangas Socio-Economic Profile

Table __. Soil Type Distribution, 2020

CITY/MUNICIPALITY	CLAY	LOAM	SANDY
1. Batangas City		x	
2. Calaca City		x	x
3. Lipa City		x	
4. Tanauan City		x	
5. Sto. Tomas City		x	
6. Agoncillo		x	
7. Alitagtag		x	
8. Balayan		x	
9. Balete	x	x	x
10. Bauan		x	
11. Calatagan	x	x	x
12. Cuenca		x	
13. Ibaan	x	x	
14. Laurel		x	
15. Lemery		x	x

C. Soil Suitability for Liberica Coffee

Soil suitability analysis plays a vital role in maximizing land production and maintaining environmental sustainability by identifying lands that are most suitable for specific crops. It provides information on the potential and constraints of an area, a prerequisite for optimum crop production. Liberica coffee requires land suitability criteria to support its growth and productivity. If an area has infertile soil conditions and has a climate that is not in accordance with the criteria for growing Liberica coffee, then several alternative solutions are needed to determine the suitability of land in that area.

The Vulnerability and Suitability Analysis (VSA), a scientific tool developed by the Bureau of Soils and Water Management (BSWM), is being used as a guide to determine the vulnerability and suitability of a particular area that could affect the production of Liberica coffee. VSA can assess the suitability of the total land area for growing Liberica coffee using five (5) agronomic factors as criteria namely: elevation, precipitation, temperature, soil pH, and soil texture.

Using VSA, the map below shows which area in the province is the most suitable for planting Liberica coffee. As shown, almost 90% of the province is suitable for growing Liberica coffee. The cities of Lipa and Tanauan, the municipalities of San Juan, Malvar, Lian, Bauan and Sta. Teresita is the most highly suitable area for such a commodity; while the least suitable areas are the municipalities of Lobo, San Luis, Lemery and Laurel.

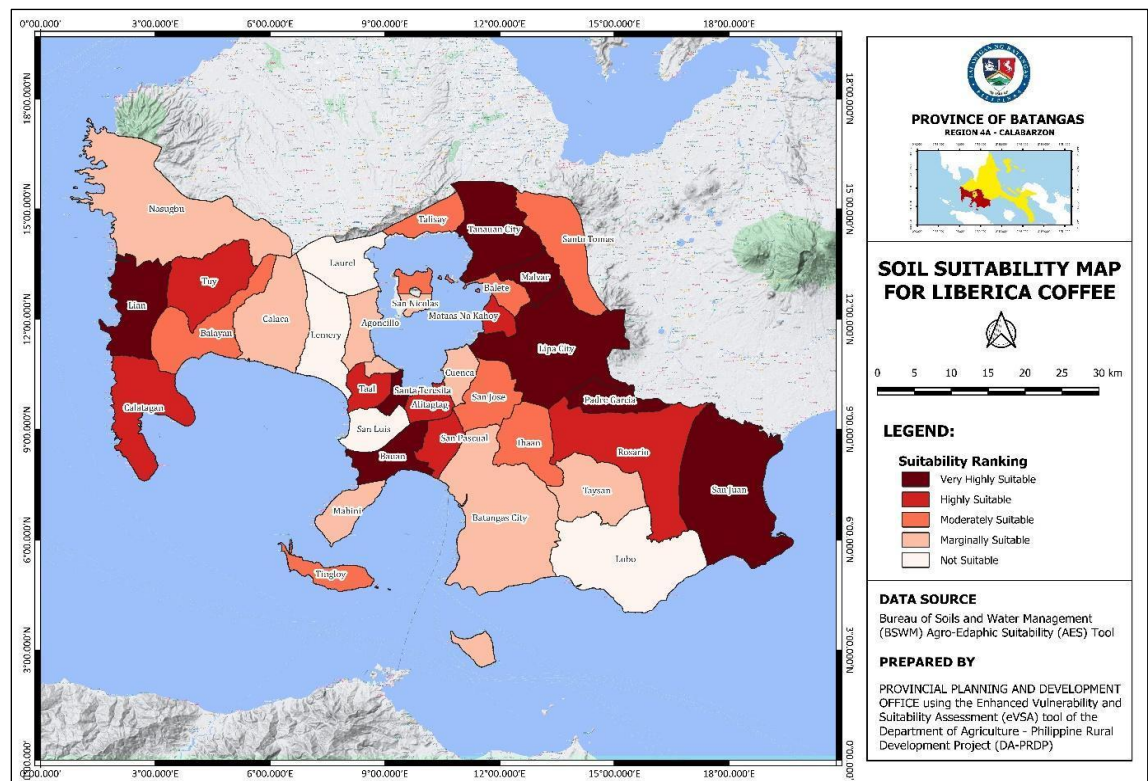


Figure __. Soil Suitability Map

D. Climate

Batangas has two types of climate – Type I and Type III (see Map 35). Type I has two seasons: dry from the months of November to April and wet during the rest of the year. Meanwhile, Type III has no very pronounced maximum rain period with a dry season lasting only from one to three months, either during the period from December to February or from March to May. This type resembles types I since it has short dry

season. Generally, the climate of the province is cool and air is humid. However, the climate varies according to one's' topography and locality. May is usually the hottest month and the coldest month is February.

The region west of the imaginary line running northeast/southwest from the foothills of Mt. Pinamucan in Batangas City and Mt. Malipunyo at the Batangas-Laguna Boundary falls under Type I. Moreover, municipalities of Agoncillo, Alitagtag, Balayan, Balete, Bauan, Calaca, Calatagan, Cuenca, Ibaan, Laurel, Lemery, Lian, Mabini, Malvar, Mataasnakahoy, Nasugbu, San Jose, San Luis, San Nicolas, San Pascual, Sta. Teresita, Taal, Talisay, Tanauan, Tingloy, Tuy and half portion of Lobo, Padre Garcia, Rosario and Taysan; and the cities of Batangas, Sto. Tomas and Lipa also falls under this type.

Verde Island and the municipality of San Juan and some portions of Lobo, Padre Garcia, Rosario and Taysan falls under Type III.

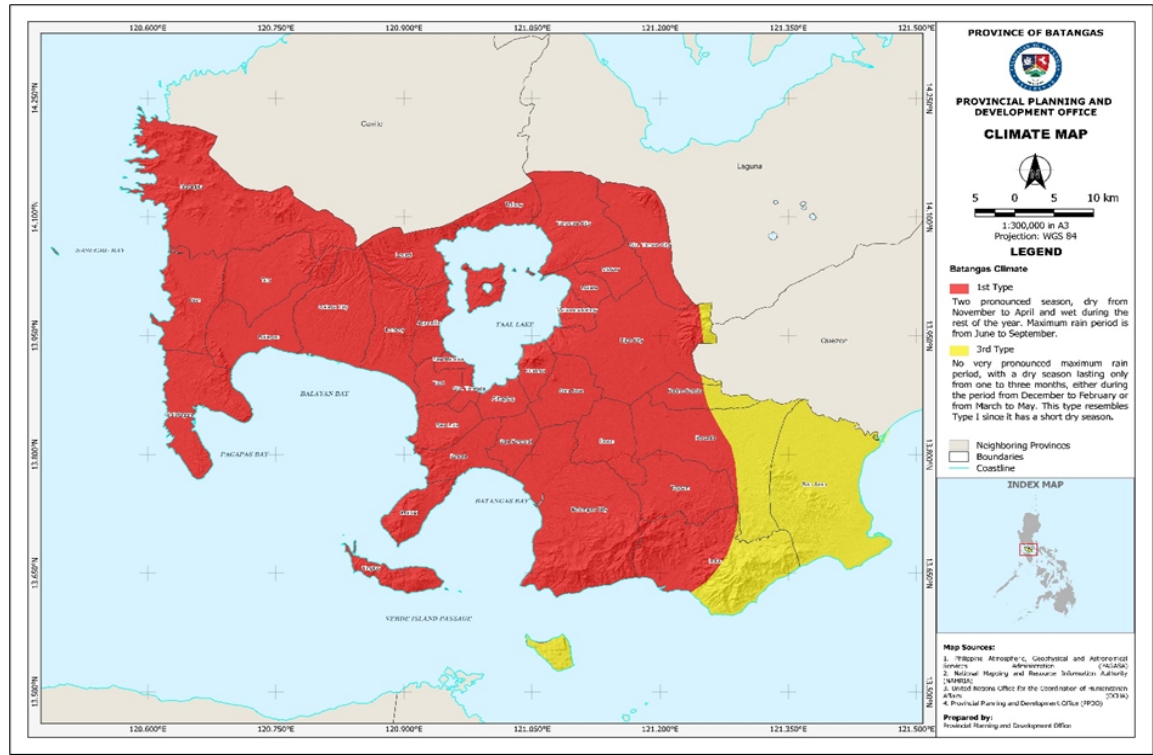


Figure __. Climate Map

Historical temperature record of the province from 1971-2000 is shown in the foregoing table:

Table __. Historical Average Temperature Record -1971 to 2000

Period (Months)	Average Temperature (in °C)
December – February	24.2
March – May	26.5
June – August	25.9
September – November	25.6

Source: Detailed Guidelines for Mainstreaming DRA and CCVA in the PDPFP, Dr. Candido A Cabrido, Jr. National Consultant, NEDA/CCA Mainstreaming Project

As projected, that there will be more frequent hot temperatures in the country with almost all areas getting warmer and the dry seasons becoming drier. There will be an increase in the number of dry days nationwide in 2020 and 2050. Accordingly, temperature of more than 35°C may be felt for 8,010 days, and within the years 2006 to 2035, there will be 6,081 dry days. However, the years 2036-2065 will see lesser dry days with 6,049 and more days with temperature of more than 35°C, numbering about 8,016.

When there is a change of temperatures, El Niño and La Niña occur. El Niño Southern Oscillation (ENSO) is a large-scale natural fluctuation of the global climate system resulting from the interactions between the ocean and atmosphere in the Central and Eastern Equatorial Pacific (CEEP). El Niño is the unusual warming in the CEEP resulting to prolonged dry season, while La Niña is the unusual cooling in the CEEP resulting to prolonged rainy season. This climate cause heavy rains in some part of the Philippines. During a period of La Niña, the sea surface temperature across the equatorial Eastern Central Pacific Ocean will be lower than normal by 3–5 °C.

Both effects of El Niño and La Niña are very important for the climate and the weather conditions in the Philippines. Simply because El Niño bring dry weather and even droughts while La Niña stands for rainy weather and floods. Impacts are on stream flow, dam operation, water allocation, domestic water supply, irrigation, hydropower allocation, depth and recharge of aquifers, water quality, watersheds and fishery. Economy, environment and communities of the four cities and 30 municipalities are all vulnerable to the variability of precipitation, frequency of typhoons, sea level rise and the risk of more droughts, floods, heat waves, forest and grassland fires brought about by the El Niño and La Niña phenomena.

Top LGUs with high exposure to flooding are the municipalities of Calatagan, Balayan and Batangas City. Calatagan having the biggest area exposure has more than half of its total land area (51 percent) or an estimated 57.092 square kilometers out of 111.97 square kilometers. The overflow of Santiago River brings threats of flooding to Barangay Lucsuhin the most. This is caused by siltation and the increase in the built-up area aggravated by high tide and extreme water pressure during heavy rains.

On the other hand, drought is an extended period when a region receives a deficiency in its water supply, whether atmospheric, surface or ground water. A drought can last for months or years, or may be declared after as few as 15 days. Generally, this occurs when a region receives consistently below average precipitation. It can have a substantial impact on the ecosystem and agriculture of the affected region. Although droughts can persist for several years, even a short, intense drought can cause significant damage and harm to the local economy.

Based on the Drought Vulnerability Index Map, municipalities that are vulnerable to drought are San Luis, Sta. Teresita, San Luis and Tingloy.

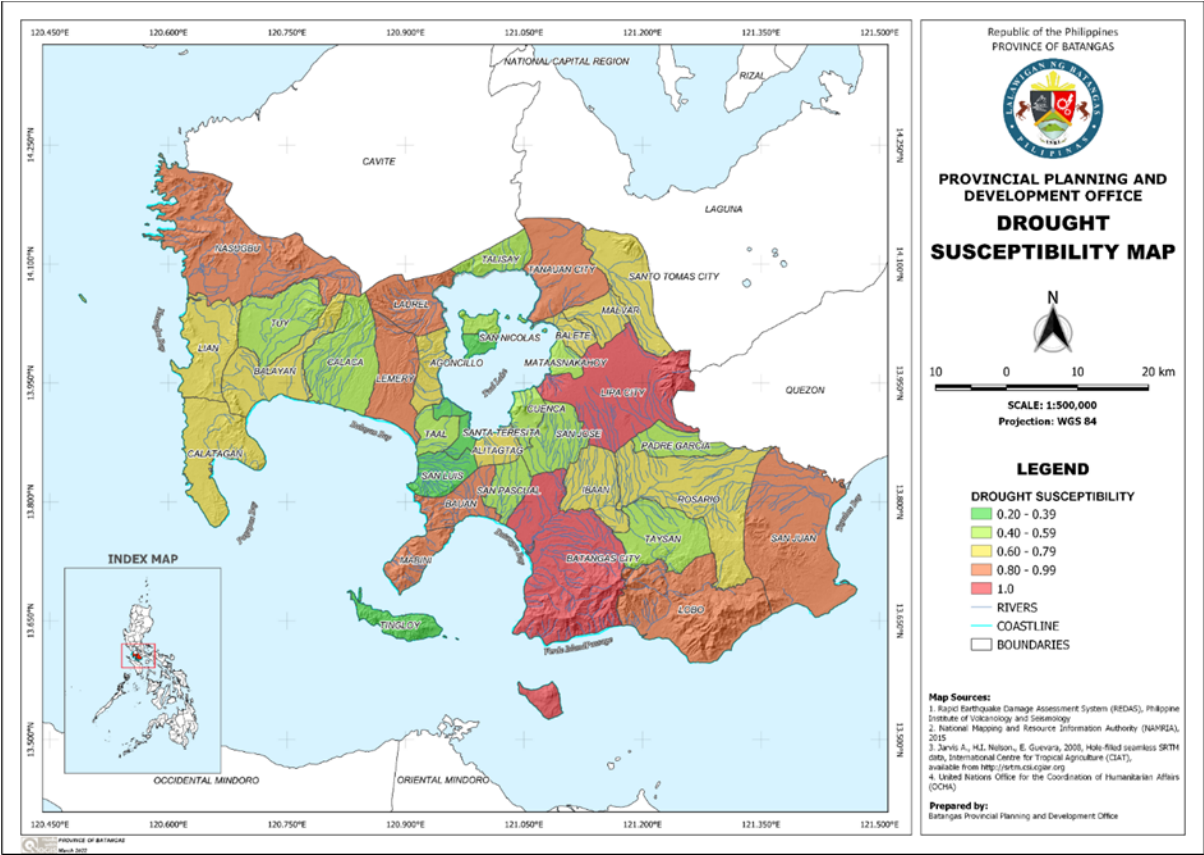


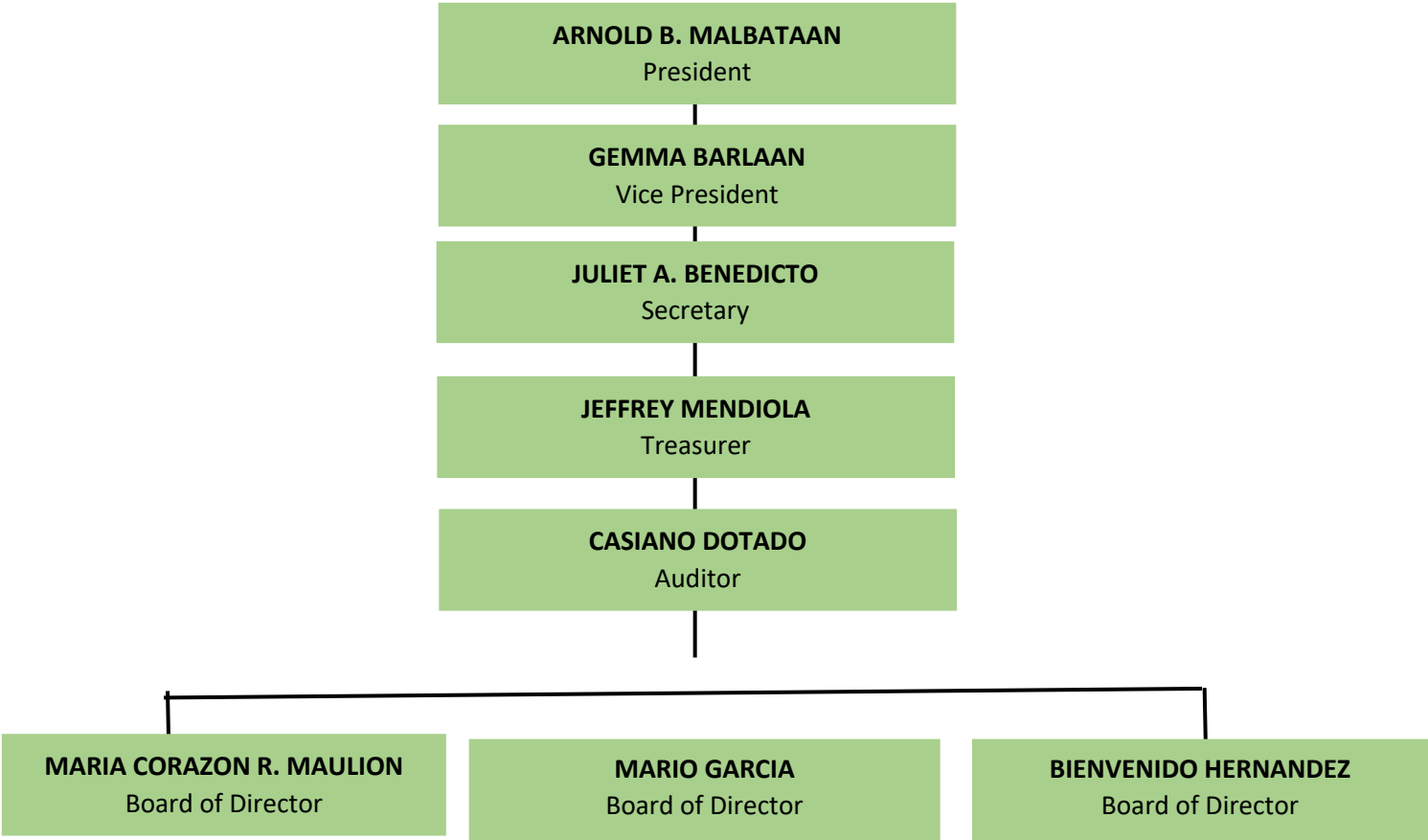
Figure __. Drought Vulnerability Index Map

E. Definition of the Geographical Area

The delimited territory (geographical area) of the Batangas Kapeng Barako encompasses the entire province of Batangas, consisting of the 5 cities and 29 municipalities.

F. Organization

The following figure shows the organizational chart of the collective group, Batangas Coffee Farmers Federation (BaCoFFed)



Part II: Technical Part

I. General Description

I.I. Morphological Characteristics

Morphological characteristics in coffee refer to the physical attributes and features of coffee plants, beans, or cherries. These characteristics can include:

I.I.I. Plant Characteristics

-Plant Height

Upright with straight trunks; grow to a height of about 9 meters

-Root System

Deep root system

-Inflorescence

Cross pollination

-Leaves

Size and Shape: Larger than those of Arabica and Robusta coffee plants. They are typically elliptic in shape, with lengths ranging from 16.4 to 35 cm (6.5 to 13.8 inches) and widths around 7.1 to 12 cm (2.8 to 4.7 inches)

Texture: Have a leathery texture, indicating they are thick and strong. The leaf blade bulges between the veins, creating a wrinkled or corrugated appearance.

Color: Dark green in color.

Margin: The leaf margin (outer edge) is wavy.

Apex: The apex (tip) of the leaf is typically acute, meaning it comes to a sharp point.

Petiole: Have a petiole, the stalk that attaches the leaf to the stem, which is also petiolate in shape.

Arrangement: Arranged opposite on the branches.

-Flowers

White, fragrant flowers are tubular and star-shaped. They are arranged in clusters near the branch tips

I.I.II. Cherry Characteristics

-Shape borne in small cluster

-oblong in shape

-thick pulp

-protruded nipple

I.I.III. Bean Characteristics

-Shape

almond shape and asymmetry

-Size

larger than arabica and robusta

-Color

tough skin and are creamy-white when raw

I.II. Harvesting Index

The Harvesting Index in coffee refers to the optimal stage of maturity at which coffee cherries should be picked to ensure the best quality and flavor. It involves evaluating the cherries' color, texture, and other characteristics to determine the ideal time for harvesting.

Common indicators of the harvesting index include:

- Color:** Changes in cherry color, such as from green to yellow, orange, red, and purple
- Firmness:** Cherries that are ripe but still firm
- Size and shape:** Fully grown and rounded cherries

I.III. Sensory (Taste/Level of Acidity)

In coffee, sensory refers to the evaluation of the coffee's taste, aroma, and overall flavor profile. Specifically, sensory evaluation assesses attributes such as:

- Flavor:** The overall taste experience, including notes of sweetness, fruit, chocolate, or other flavor profiles.
- Acidity:** The bright, lively, or sharp sensation in the coffee, which can range from low to high.
- Body:** The texture or mouthfeel of the coffee, which can range from light to heavy.

Sensory evaluation involves tasting and assessing the coffee's:

- Flavor notes (e.g., fruity, floral, nutty, woody, smoky, earthy)
- Acidity level (e.g., bright, balanced, low)
- Body (e.g., light, medium, heavy)
- Balance and overall impression

I.IV. Genetic Profile

According to the summary report of the BARAKO Project, a collaborative effort between Batangas State University and De La Salle University's Practical Genomics Laboratory Center for Natural Sciences and Environmental Research (CENSER), funded by the Department of Agriculture-Bureau of Agricultural Research, a molecular identification study was conducted on *Coffea* parent trees in Batangas, Philippines. The study analyzed 52 coffee samples and found that 50 out of 52 samples belonged to the *Coffea liberica* species, based on DNA extraction, PCR amplification, and Agarose Gel Electrophoresis (AGE) of the *matK* gene. This conclusion was supported by the alignment of 143 sequences from the Batangas samples and NCBI Genbank sequences.

The 50 *Coffea liberica* samples identified in the molecular identification study will serve as a baseline for compliance in verifying the authenticity of Liberica coffee for certification purposes, ensuring that coffee products labeled as Liberica meet specific genetic standards.

I.V. Traceability

The Local Government Unit (LGU), through its Agriculturist office, will issue a certificate of locality, verifying that the coffee trees are grown within their specific geographic area, thereby supporting traceability and authenticity of the Liberica coffee produced in the province.

II. Specific Description of the product

II.I Roasted Coffee Beans

Aroma (Specific for liberica)

Degree of Roast (It could be light, medium or dark roast)

II.I Ground Coffee

Aroma (Specific for liberica)

Taste/Body (Bitterness/Sweetness)

Acidity (What is the range of acidity)

Degree of Roast (It could be light, medium or dark roast)

III. The Production Process

III.I. Newly established plantation

III.I.I. Recommended requirements

III.I.I.I. Before planting

Planting materials

Planting materials or seedlings should be free from pests and diseases.

For the establishment of new farms, planting materials/varieties registered at the Bureau of Plant Industry (BPI) - National Seed Industry Council (NSIC) are recommended.

Likewise, quality planting materials/seedlings are recommended to be sourced from the BPI certified plant material/seedlings coming from the accredited plant nursery operators. Records of the source of planting materials shall be kept.

The source of planting materials should be from identified and tagged mother trees by the Provincial Technical Working Group for Coffee.

It could be propagated from seeds, grafting, cutting (clonal) and tissue culture.

III.I.I.II. When planting (newly established farm)

Soil requirements

- Minimum soil depth is 1.5 meter
- Optimum soil ph is between 5.5 and 6.5
- Highly fertile with high moisture holding capacity
- Medium texture with good drainage and erosion

Soil Conservation

For areas with steep slope (above 18%), coffee plantations should implement erosion control measures which include contouring, silt pits, mulching, cover cropping and planting crops against the slope. In planting coffee trees along the contour, the rows of coffee should run at right angles to the direction of the slope. Meanwhile, other anti-erosion measures such as contour ridges, contour bunds, contour ditches and vegetative measures that are designed to disrupt the downward flow of rainfall run-off, collect, and convey the run-off to major waterways and then to a drainage channel should also run along the contour between the coffee rows. The examination of the contour map and the preparation of outline plans for selected erosion control measures will determine the contour interval at which the rows of coffee are to be planted.

Lay-outing

Establish a straight line along the boundary. Rows are from east to west or north to south and allow sunlight to penetrate between rows. Recommended planting distance is 2m x 2m. If intercropping is desired, the recommended planting distance is 4m x 4m.

Staking and Holing

- Mark the area to place the planting holes by staking the rows and the hills.
- Dig holes of around 40cm x 40cm x 40cm.
- Separate topsoil from subsoil when digging.
- Add compost and fertilizers rich in phosphorus to the holes, refill by putting back the topsoil first then the subsoil.

Transplanting

- Field planting can begin when the coffee seedlings in bags have a minimum of six to eight leaf pairs.
- Planting out in the field should be done on-set of rainy seasons and on cloudy days.
- Avoid planting trees when conditions are windy, hot and dry, or during the hottest part of the day.
- Before planting, thoroughly water the seedlings in the growing bags.
- Remove the plastic bag before planting.
- Place the seedling upright in the hole; do not plant at an angle. Make sure that the root collar is exactly at ground level to prevent rotting.
- Compact the base around the seedling.
- Basal fertilization is recommended using complete fertilizer. Apply basal fertilizer 75 grams seedlings.

Weeding

Weeds compete for both nutrients and water, so it is essential to keep the area under the canopy of the trees, weed-free.

Weeding should be done at least three to four times per year, especially in the wet season.

Mulching

To maintain soil moisture and control weeds, coffee plants should be mulched with rice straw or other appropriate material to a depth of 2 to 3 inches (5 to 8 cm) especially at the end of the wet season.

III.I.I.III. Care and Maintenance (Non-bearing and Bearing)

Weeding

Regular weeding should be done to avoid competition on nutrient uptake and to avoid incidence of pests and diseases.

Fertilization

Growers or agricultural workers must not use untreated solid nor liquid manure because pathogenic microorganisms can persist in soils for long periods of time.

In cases when the farm produces its own organic inputs (e.g. compost), proper treatment procedures should be adopted to reduce or eliminate the pathogens present in the raw material and to minimize the probability of contaminating the product. A record of treatment procedures, including the raw materials used should be kept. The location of the composting site should also consider the slope and its proximity to production areas in order to prevent cross contamination from run-off or leaching. Composting area should be located at the lowest part of the production area. Records of agricultural inputs should include origin and composition of fertilizers, date, frequency and location of application. Human manure and urine are not allowed.

In order to optimize yield, considerations like crop demand and available nutrients from the soil, farm manure and other crop residues may be useful in deciding the appropriate kind of fertilizer or amount of application.

Soil analysis is highly recommended as a basis for the application of fertilizer. In the absence of soil analysis, sample computation of the recommended rate of fertilizer application per tree. If organic fertilizer is preferred, this should be equivalently computed according to the sample computation.

Growers should use only registered agricultural chemicals and should use these for its intended purpose and according to manufacturer's instructions. All organic and inorganic fertilizers are to be used appropriately, optimizing yield and minimizing negative impacts on human health, the environment and the quality of the coffee.

Fertilizers must be clearly labeled and should be stored in a way that does not contaminate the environment.

Water management

If coffee plantations are rainfed, plantings should be done at the onset of the rainy season. Otherwise, irrigation is recommended for planting during dry months. Care should be taken to ensure that soil around the coffee tree

is not dried out after planting and that the moisture is maintained. Furthermore, water use should be managed to ensure sustainability.

Removal of Vertical Sprouts/Desuckering

Remove vertical sprouts that grow on the main trunk regularly, these absorb more water and nutrients than the lateral branches.

Detopping or Topping

Described as cutting back shoots of coffee trees to keep it at less than 1 to 1.5 meters high. This operation is done to newly planted or newly rejuvenated coffee trees. Detopped coffee trees give dwarf appearance, encourage elongation of branches and production of more twigs, ultimately leading to higher yield.

Maintenance Pruning

Pruning will increase, maximize and improve yield, as such is a necessary activity for the growers. Pruning is recommended to maintain the correct balance between leaf area and crop, thus, preventing overbearing and dieback, maintaining good tree shape, reducing biennial bearing and providing a supply of good growth the next season's crop.

1. Single Stem Pruning

- YEAR 1: Desucker to maintain a single stem and avoid competition
- YEAR 2: Desucker to remove drooping primary branches that touch the ground. Cut back to the nearest secondary branch.
- YEAR 3: Trees should be allowed to crop in the third year.
Cap the main stem by cutting above a side primary shoot at about 5 ft (1.6 m) from the ground. Remove secondary branches within 8 inches (20 cm) of the main stem.
Maintain a maximum number of wellspaced secondary branches on each primary branch.
Remove all dead, weak and pest or disease damaged branches.

2. Multiple Stem Pruning

- Bend down the first main stem and tie it to a stake. Let three to five vertical sprouts develop.
- Cut off the bent stem when the vertical sprouts grow up to 30cm and select only three (3) healthy sprouts to grow as the main stems.
- Spread the new three (3) main stems using a bamboo expander, this will give even distribution of sunlight to the trees.

Rejuvenation

Coffee rejuvenation is the cutting of vertical stems or trunks of old coffee trees to induce growth of new sprouts. Rejuvenation brings back 100% of the coffee green bean yield and reduces labor cost up to 50%. The ideal age of tree to be rejuvenated normally ranges 10 to 30 years depending on the tree vigour and yield pattern. Grafting may be employed to improve coffee productivity.

Two rejuvenation methods:

1. Side Pruning

- Involves removing one side of the tree, training a new sucker and then removing the other side of the tree two years later.
- This is recommended for all growers, as only 50% of the crop is lost for the two-year period.
- Two years before stumping, remove all branches on the eastern side of the tree after harvesting.
- Select a new sucker approximately 12 to 18 inches (30-45 cm) from the ground, and train the shoot until bearing a crop.
- Two years later, stump the older stem above the new stem.
- Cut a 45 degree angle. Do not cut straight.

2. Full Stumping

Cutting back the tree to knee height – about 12 inches (30 cm) from the ground level, and to develop a new stem from the stump.

Rehabilitation

Involves wedding, proper fertilization, desuckering, pruning, and proper management of pests and diseases to develop a new productive tree.

Pest and Disease Management

In order to control infestation and diseases such as coffee stem borers, coffee berry borer, coffee rust and berry rot, the grower should regularly inspect and monitor the production site for the presence of pests and diseases. Once observed, the infected tree(s) should be tagged and recorded for further observations. The results of the inspection and monitoring will be used as a basis for deciding the best management strategy (i.e. management or control) and implementing interventions to treat the affected area(s), thus, eradicating pests and/or diseases. Aside from the production area, other facilities are also checked for presence and population density of pests, and the damages it has caused.

Integrated Pest Management (IPM) is an effective and environment-friendly approach to pest management to control and minimize pest damages. IPM combines the use of current and comprehensive information on the life cycles of pests; their interaction with the environment and the available pest control methods, e.g. varietal selection, biological, cultural, physical, mechanical and chemical controls. Use of chemicals should be on a need basis.

Use of Biological control agents

Environmental and consumer safety should be considered when using competing biological organisms and/or their metabolites to control pests, mites, plant pathogens and spoilage organisms in coffee production. Growers should only use biological control agents, which are authorized for the cultivation of coffee and should use them for its intended purpose and according to the manufacturer's instructions. Registration of biological

control agents is based on existing national standards and regulations of BAFS.

Use of pesticides

The harvested coffee cherries are usually protected using chemical crop protection products. Pesticides that are target-specific and have minimal negative effect/impact on the agro-ecosystem and environment should be used. Choice of crop protection strategies should be appropriate to the pests and diseases being controlled. Growers should use agricultural chemicals that are FPA-registered for coffee cultivation and/or approved in the country where the produce is to be traded.

These should be procured from licensed suppliers. Such agricultural chemicals must be used according to its intended purpose, approved label instructions and applied at approved dosages to prevent levels exceeding the maximum residue limits (MRLs). Growers who apply agricultural chemicals should be trained on proper application.

Pesticides must be always clearly labeled and stored in the original container. They should be stored in a designated and secured area away from the main production area. Pesticides and other agrochemicals must be stored separately in an elevated area using pallets or similar material.

III.I.I.IV. Harvesting operations

Harvesting

Normally, coffee cherries are harvested according to maturity indices to ensure good quality coffee. The harvesting is usually done by selective picking of ripe coffee cherries, which are ready for harvest within 7 to 14 months after flowering depending on the species and location.

In general, harvesting is done by any of the two (2) basic systems such as: stripping - where cherries harvested are predominantly ripe, and selective picking (hand picking) - where only ripe cherries are harvested. Of the two (2) systems, selective picking of ripe cherries is highly recommended.

During harvesting, coffee cherries should be placed in clean harvesting containers prior to segregation or selection operations. Harvested cherries should not be placed in direct contact with the ground. Furthermore, newly harvested cherries should not be mixed with any of the fermented part of previously harvested cherries (e.g. from the previous day) as this will contaminate the current batch of coffee cherries and may result in deterioration of the entire batch.

Harvested cherries that are heavily bruised, damaged, diseased, or over-ripe should be segregated. Those that cannot be made safe by further processing should be disposed of properly to avoid contamination.

Containers (e.g. basket and plastic containers) used for harvesting should be made from non-toxic materials. These containers should be designed and constructed to ensure that these can be cleaned, disinfected and maintained in working condition to avoid contamination. When using

reusable harvesting containers and tools, a cleaning and disinfection schedule should be in place to prevent contamination. A record of cleaning should be available. Containers that are damaged and can no longer be kept in a hygienic condition should be discarded.

As far as practicable, harvesting containers should only be used to contain harvested produce. If these containers are used for other purposes, these must be cleaned and disinfected as necessary prior to use. Containers of agrochemicals shall not be used as harvesting containers.

III.I.I.V. Post-harvest practices

This section applies hygienic recommendations for the primary processing of harvested ripe coffee cherries, which will then be supplied to processors for further processing. It provides recommendations on the application of relevant Good Manufacturing Practices (GMPs) principles for all stages of primary processing of coffee beans.

Types of Coffee Processing

Dry Method

- Oldest/simplest/cheapest method
- Laid out on the ground (cemented floor or concrete pavement)
- Stir or rake regularly (2-3 times a day)
- Takes place anywhere from 10 days to 3 weeks
- “Dry-processed”, “unwashed”, or “natural coffee”

Wet Method

- Involves more capital, more water and more care
- The pulp is removed from the berries within 12-24 hours after harvesting
- Berries are soaked in a large tank of water to separate unwanted berries
- 18-24 hours fermentation
- The coffee is thoroughly wash with clean water to remove the remaining parchment
- “wet-processed”, “washed”, or “mild coffee”

Semi-dry Method

- A hybrid process which the outer skin is removed as in wet process, but the sticky fruit residue (mucilage) is allowed to dry on the bean
- “wet-hulled”, “semi-washed coffee”

Drying

Start within 12 hours after harvest to ensure high quality green coffee

Over-dried - becomes brittle and produce too many broken during hulling

Too moist - prone to rapid deterioration caused by the attack of microorganisms

Fully-dried - when the beans produce sound whenever being shaken by hands

Drying Systems

Sun Drying

- Coffee cherries are laid out on cemented floor
- The operation takes place from 10 days to 3 weeks
- The moisture is reduced to 12-14%
- 3 weeks for Excelsa and Liberica

Solar Drying

- Coffee cherries are placed in trays of solar drier
- Layer should be no deeper than one inch

Hulling

removal of the outer coverings of the bean

Polishing

used to remove any silver skin and parchment like husk

Cleaning and Sorting

Selection or taking out of undesirable foreign materials or broken pieces of coffee beans (Annex)

Grading

Process of categorizing coffee beans on the basis of criteria such as size of beans and the number of defects or imperfection in a given sample of green coffee beans

III.I.I.VI. Roasting

Roasting

- It is chemical process by which aromatics, acids, and other flavor components are either created, balanced, or altered in a way that should augment the flavor, acidity, aftertaste and body of the coffee
- Done using roaster
- Length of time and temperature are the most crucial factors (200-225 C temperature & 15-20 min time)
- The beans lose a good amount of moisture (20-22%) and about 10-15% of the bean's caffeine is lost
- When the desired roast has been reached, the beans are then quickly dumped into a cooling drum with mixing arms

Degree of Roast

- Light Roast
- Medium Roast
- Dark Roast

III.I.I.VII. Grinding

Exposes the beans' cell wall and prepares the beans to release their flavor during brewing

Degree of Grinding

- Course
- Medium
- Fine
- Extra Fine
- Turkish

III.I.I.VIII. Packaging

Whole roasted coffee beans or freshly-ground coffee is packaged in valve-sealed bags or vacuum-sealed bags. The minimum packaging is 150 grams while the maximum is 1000 grams.

III.I.I.IX. Storage

If storage is necessary, the bagged green coffee beans must be placed in pallets or similar material so that it will not be in direct contact with the ground; and in storage sheds that are weather-proof, well ventilated, free from moisture and insect pests and located away from sources of smoke and other odours that may contaminate the coffee. Storage facilities should be designed to minimize damage to the coffee. Moisture content of the green coffee beans should be monitored regularly during storage and maintained at a maximum of 12 percent. Storage area should be kept clean to avoid pest infestation.

Part III. Control and Traceability

III.I. Internal

All coffee growers should be registered to Batangas Coffee Farmers Federation (BaCoFFed).

III.II. Traceability

Provincial Technical Working Group and Batangas Coffee Farmers Federation will maintain the records and control compliant to Code of Practice.

Part IV. Labelling

The Batangas Kapeng Barako will have a distinct seal. Only producers who comply with the specifications contained in the Code of Practice shall be allowed to use the seal.

ANNEX

Table __. Sample numbers, their corresponding tree codes in Batangas, and their respective groups based on the *matk* gene. **Indicate parent trees that clustered with the Canephora group.

Sample Number	Tree Code	Group cluster <i>matk</i> gene	Sample Number	Tree Code	ID based on <i>matk</i> gene
1	KBP-001 AF	Liberica group	27	ELB004	Liberica group
2	KBP-002 AF	Liberica group	28	ER001	**Canephora group
3	KBP-003 AF	Liberica group	29	ESLB002	Liberica group
4	KBP-004 AF	Liberica group	30	ESLB003	Liberica group
5	KBP-005 AF	Liberica group	31	ESLB004	Liberica group
6	KBP-006 AF	Liberica group	32	ESLB005	Liberica group
7	KBP-007 AF	Liberica group	33	JFEX001	Liberica group
8	KBP-008 AF	Liberica group	34	JFLB001	Liberica group
9	KBP-009 AF	Liberica group	35	JFLB002	Liberica group
10	KBP-010 AF	Liberica group	36	JFLB003	Liberica group
11	AGLB003	Liberica group	37	JFLB004	Liberica group
12	AGLB004	Liberica group	38	Ma1	Liberica group
13	AGLB005	Liberica group	39	OPA1	Liberica group
14	AGLB006	Liberica group	40	OPA3	Liberica group
15	AGLB007	Liberica group	41	PTLB001	Liberica group
16	AGLB008	Liberica group	42	PTLB002	Liberica group
17	AGLB009	Liberica group	43	PTLB003	Liberica group
18	Ba1	Liberica group	44	PTLB004	Liberica group
19	Ba2	Liberica group	45	PTLB005	Liberica group
20	BFEX001	**Canephora group	46	Sa1	Liberica group
21	BFEX002	Liberica group	47	STLB01	Liberica group
22	BFLB001	Liberica group	48	STLB02	Liberica group
23	BFLB002	Liberica group	49	Ta1	Liberica group
24	BT001	Liberica group	50	TLLB001	Liberica group
25	ELB002	Liberica group	51	TLLB002	Liberica group
26	ELB003	Liberica group	52	TLLB003	Liberica group

Sample Computation

Generally, the table below shows a sample computation of the recommended rate of fertilizer application per tree.

Table __. Sample combination of fertilizer materials to satisfy the fertilizer recommendation (120-120-60) for non-bearing coffee trees.

Year 1 (Non Bearing)	Schedule Every 3-months			
TYPE OF FERTILIZER	1 st Application	2 nd Application	3 rd Application	4 th Application
1) <i>Granular (commercial)</i>	grams/tree	grams/tree	grams/tree	grams/tree
Diammonium phosphate (18-46-0)	15	30	50	60
Urea (46-0-0)	9	18	30	36
Muriate of Potash (0-0-60)	6	12	20	24
<i>Total granular fertilizer per tree</i>	30	60	100	120
2) <i>Organic Fertilizer (as defined by the PNS) application per tree</i>	At least 1kg/tree	At least 1kg/tree	At least 1kg/tree	at least 1kg/tree
3) <i>Foliar Fertilizer</i>	based on product label recommendations and soil analysis			

Table __. Sample combination of fertilizer materials to satisfy the fertilizer recommendation (120-60-120) for bearing coffee trees.

Year 2-10 (Bearing)	Schedule Every 4-months		
TYPE OF FERTILIZER	1 st Application	2 nd Application	3 rd Application
1) Granular (commercial)	grams/tree	grams/tree	grams/tree
Diammonium phosphate (18-46-0)	29	29	34
Urea (46-0-0)	38	38	43
Muriate of Potash (0-0-60)	38	38	43
Total granular fertilizer per tree	105	105	120
2) Organic Fertilizer application per tree	At least 1kg/tree	at least 1kg/tree	At least 1kg/tree
3) Foliar Fertilizer	based on product label recommendations and soil analysis		

ANNEX

List of Green Coffee Defects

Black beans	Brown to black beans Shrunken Wrinkled Flat faced Crack too opened	Lack of water during development of cherries Over fermentation Overripe cherries picked from the ground	
Moldy beans	Coffee beans infested by mold presenting yellowish or reddish spores	Over fermentation Long interruptions during drying process Storage of beans with high moisture content	
Sour or Partial Sour beans	Light brown to dark brown beans Crack free of tegument Silver skin can be reddish brown	Delay in between coffee picking and depulping Overextended fermentation process Use of dirty water Storage of beans with high moisture content	
Crystallized beans	Coffee beans grayish and bluish in color, fragile, brittle	Beans dried mechanically at temperatures above 60° Celsius, faulty in drying process	

Faded or Streaked beans	Stained bean, showing an irregular greenish color	Usually due to uneven drying or re-wetting after drying	
Faded-Oldish beans	Coffee beans with alterations in the normal color, white coffee beans, cream, yellow to brown	Long time in storage Storage in bad conditions (high temperature and humidity) 25-30 C and 50-60% would be ideal	
Faded-Amber or Buttery beans	Yellow, semi transparent beans	Iron deficiency in the soil	
Faded-Overdried beans	Amber to yellow beans	Excessive time in the dryer, or high temperature during drying	
Faded-Cut or Nipped beans	Cut or nipped bean and rusty	Bean broken by the depulping machine during wet process Cherries harvested prior to maturity	

Insect-Damaged beans	Beans with small holes caused by insects.	Attack on cherries by <i>Hypothenemus haempei</i> (coffee berry borer) Attack by <i>Araecerus Fasciculatus</i> during storage, due to inadequate storage conditions	
Shrunk beans	Wrinkled bean	Beans affected by drought conditions Lack of fertilization	
Immature beans	Green or light grey beans Beans tend to have a very adherent silver skin and size smaller than normal Whither surface Smaller size than normal In this group low density beans are included		
Pressed or Crushed beans	Bruised beans with partial fractures	Bruised beans during depulping, drying, and milling process	

		Milling parchment with high moisture	
Wet or Undried beans	Beans with dark green color and soft texture	Bean with high humidity Lack or incomplete drying	

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